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M. B. A. (Third Semester) Examination, Nov.-Dec. 2017

(New Course)

(Management Branch)

(Specialization : General)

OPTIMIZATION METHODS

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 32

Note *: All units are compulsory. Attempt any two questions from each unit. Each question carries 8 marks.*

Unit - I

1. (a) The standard weight of a special purpose brick is 5 kg. and its contains two basic ingredients B_1 and B_2. B_1 costs Rs. 5/kg and B_2 cost Rs. 8/kg. Strength

considerations dictate that the brick contains not more than 4 kg of B_1 and a minimum of 2 kg of B_2. Since the demand for the product is likely to be related to the price of the brick, find graphically the minimum cost of the brick satisfying the above conditions.

(b) Solve the following LPP using Simplex method :

Max $z = 2x_1 + 5x_2$

Subject to

$$x_1 + 2x_2 \leq 8$$

$$x_1 \leq 4$$

$$0 \leq x_2 \leq 3$$

x_1 unrestricted

(c) How does Simplex algorithm indicate that :

(i) there is an alternate optimal solution?

(ii) the problem has unbounded optional solution?

(iii) the problem has no feasible solution?

Unit - II

2. (a) A product is produced by four factories A, B, C and D. The unit production cost in them are Rs. 2, Rs. 3. Rs. 1 and Rs.5 respectively. Their production capacities are : factory A = 50 units, B = 70 units, C = 30 units and D = 50 units. These factories supply the product to four stores, demands of which are 25, 35, 105 and 20 units respectively. Unit transport cost in rupees from each factory to each store in given in the table below :

		Stores			
		1	2	3	4
Factories	A	2	4	6	11
	B	10	8	7	5
	C	13	3	9	12
	D	4	6	8	3

Determine the extent of deliveries from each of the factories to each of the stores so that the total production and transportation cost is minimum.

(b) Priyanshu Enterprise has three factories at locations A, B and C which supplies three warehouses located at D, E and F. Monthly factory capacities

are 10, 18 and 15 units respectively. Monthly warehouse requirements are 75, 20 and 50 units, respectively. Unit shipping cost (in Rs.) are given here :

	Warehouse		
Factory	D	E	F
A	5	1	7
B	6	4	6
C	3	2	5

The penalty cost for not satisfying demand at the warehouse D, E and F are Rs. 5, Rs. 3 and Rs. 2 per unit respectively. Determine the optimal distribution for Priyanshu using transportation method.

(c) A company manufacturing air coolers has two plants located at Mumbai and Kolkata with a capacity of 200 units and 100 units per week respectively. The company supplies the air-coolers to its four show rooms situated at Ranchi, Delhi, Lucknow and Kanpur which have a maximum demand of 75, 100, 100 and 30 units respectively. Due to the differences in raw material cost and transportation

cost. The profit per unit in rupees differs which is shown in the table below :

	Ranchi	Delhi	Lucknow	Kanpur
Mumbai	90	90	100	110
Kolkata	50	70	130	85

Plan the production programme so as to maximize the profit. The company may have its production capacity at both plants partly or wholly unused.

Unit - III

3. (a) A company has four territories open and four salesman available for assignment. The territories are not equally rich in their sales potential. It is estimated that a typical salesman operating in each territory would bring in the following annual sales:

Territory :	I	II	III	IV
Annual Sales (Rs.) :	60000	50000	40000	30000

The four salesman are also considered to differ in ability; it is estimated that working the same conditions, their yearly sales would be proportionatary as follows :

Salesman :	A	B	C	D
Proportion :	7	5	5	4

If the criterion is maximum expected total sales, the intuitive answer is the asign the best salesman to the richest territory, the next best salesman to the second richest territory and so on. Verify this answer by the assignment method.

(b) A firm marketing a product has four salesmen S_1, S_2, S_3 and S_4. There are three customers C_1, C_2 and C_3. The probability of making a sale to a customer depends upon the salesman customer support. The table below respresents the probability with which each of the salesman can sell to each of the customers :

		Salesmen			
		S_1	S_2	S_3	S_4
	C_1	0.7	0.4	0.5	0.8
Customer	C_2	0.5	0.8	0.6	0.7
	C_3	0.3	0.9	0.6	0.2

If only are salesman is to be assigned to one customer, what combination of salesman and customers shall be optimal? Profit obtained by selling one unit to C_1 is Rs. 500, to C_2 is Rs. 450 and to C_3 is Rs. 540. What is the total expected profit?

(c) A salesman wants to visit cities 1, 2, 3 & 4. He does not want to visit any city twice before completing the tour of all the cities and wishes to his home city, the starting station. Cost of going from one city to another in rupees is given in the table below. Find the least cost route :

		To city			
		1	2	3	4
	1	–	30	80	50
From city	2	40	–	140	30
	3	40	50	–	20
	4	70	80	130	–

Unit - IV

4. (a) Give a general structure of queuing system and explain. Illustrate some queuing situation?

(b) Arrivals at telephone booth are considered to be poisson with an average time of 10 minutes between one arrival and the next. The length of phone call is assumed to be distributed exponentially, with mean 3 minutes :

(i) What is the probability that a person arriving at the booth will have to wait?

(ii) What is the average length of the queue that forms time to time.

(c) A self service store employs one cashier at its counter. Nine customers arrive on an average every 5 minutes while the cashier can serve 10 customer in 5 minutes. Assuming poisson distribution for arrival rate and exponential distribution for service time, find :

(i) Average no. of customers in the system.

(ii) Average no. of customers in the queue.

(iii) Average time a customer spends in the system.

(iv) Average time a customer waits before being served.

Unit - V

5. (a) Listed in the table are the activities and sequencing necessary for maintenance job on the heat exchange in a refinery :

Activity	Description	Predecessor activity
A	Dismantle pipe connections	—
B	Dismantle heater, closure and floating front	A
C	Remove tube bundle	B
D	Clean bolts	B
E	Clean heater and floating head front	B
F	Clean tube bundle	C
G	Clean shell	C
H	Replace tube bundle	F, G
I	Prepare shell pressure test	D, E, H
J	Prepare tube pressure test and reassemble	I

Draw a network diagram for the project.

(b) The following table shown for each activity needed to complete the project the normal time, the shortest time in which the activity can be completed of a building contract and the cost per day for reducing

the time of each activity. The contract includes a penalty clause of Rs. 100 per day over 17 days. The overhead cost per day is Rs. 160.

Activity	Normal Time in days	Shortest Time in days	Cost of Reduction per day (Rs.)
1-2	6	4	80
1-3	8	4	90
1-4	5	3	30
2-4	3	3	—
2-5	5	3	40
3-6	12	8	200
4-6	8	5	50
5-6	6	6	—

The cost of completing the eight activities in normal time is Rs. 6500.

(x) Calculate the normal duration of the project, its cost and the critical path.

(y) Calculate and plot on a graph and Cost/Time function for the project and state :

(i) the lowest cost and associated time.

(ii) the shortest time and associated cost.

(c) Write short notes on the following :

(i) Dummy acitivity

(ii) Merge and Burst events

(iii) Total float and free float

(iv) Critical activity and Critical path